

Dataset specifications

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Geospatial dataset and methods described in manuscript:

Scheer, J.; Caduff, R.; How, P.; Marcer, M.; Strozzi, T.; Bartsch, A.; Ingeman-Nielsen, T. Thaw-Season InSAR Surface Displacements and Frost Susceptibility Mapping to Support Community-Scale Planning in Ilulissat, West Greenland. *Remote Sens.* 2023, 15, 3310. <https://doi.org/10.3390/rs15133310>

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1 DATASET DESCRIPTION

The dataset contains one ZIP-archive folder (Frost_suscept_mapping_Ilulissat_Greenland.zip) containing tiff rasters derived from the processing of remotely-sensed synthetic aperture radar and optical images, for the area of Ilulissat (West Greenland). The folder also includes a TAB-delimited text file with the ground truth data (active layer thicknesses and floristic compositions) that were used as inputs and validation during the generation of the rasters.

2 RASTER SPECIFICATIONS

2.1 Yearly thaw-season ground surface displacements

Table 1. Line-of-sight (LOS) thaw-season ground surface displacement raster – year 2015.

Attribute	Description
File name	ILU_InSAR_LOS_surface_displacement_2015
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site Extent	Ilulissat, West Greenland Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width Height Pixel size	680 pixels 885 pixels 10 m
Band(s)	1
Raster description	2015 thaw-season ground surface displacement raster (unit m), provided in the radar line-of-sight (LOS) direction.
Derived from (raw data): Acquisition period Track Mode Incidence angle Revisit Reference point Digital Elevation Model	Sentinel-1 (Level-1 Single Look Complex) data from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31). 2015-07-26 to 2015-09-12 Ascending 90 IW 52.2 ° 12 day Located on bedrock (69°13'48.37" N, 51°0'46.21" W) Arctic DEM (10 m) (Porter et al., 2018)

Table 2. Line-of-sight (LOS) thaw-season ground surface displacement raster – year 2016.

Attribute	Description
File name	ILU_InSAR_LOS_surface_displacement_2016
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	2016 thaw-season ground surface displacement raster (unit m), provided in the radar line-of-sight (LOS) direction.
Derived from (raw data):	Sentinel-1 (Level-1 Single Look Complex) data from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31).
Acquisition period	2016-06-02 to 2016-09-18
Track	Ascending 90
Mode	IW
Incidence angle	52.2 °
Revisit	12 day
Reference point	Located on bedrock (69°13'48.37" N, 51°0'46.21" W)
Digital Elevation Model	Arctic DEM (10 m) (Porter et al., 2018)

Table 3. Line-of-sight (LOS) thaw-season ground surface displacement raster – year 2017.

Attribute	Description
File name	ILU_InSAR_LOS_surface_displacement_2017
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	2017 thaw-season ground surface displacement raster (unit m), provided in the radar line-of-sight (LOS) direction.
Derived from (raw data):	Sentinel-1 (Level-1 Single Look Complex) data from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31).
Acquisition period	2017-06-03 to 2017-09-17
Track	Ascending 90
Mode	IW
Incidence angle	52.2 °
Revisit	6 day
Reference point	Located on bedrock (69°13'48.37" N, 51°0'46.21" W)
Digital Elevation Model	Arctic DEM (10 m) (Porter et al., 2018)

Table 4. Line-of-sight (LOS) thaw-season ground surface displacement raster – year 2018.

Attribute	Description
File name	ILU_InSAR_LOS_surface_displacement_2018
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	2018 thaw-season ground surface displacement raster (unit m), provided in the radar line-of-sight (LOS) direction.
Derived from (raw data):	Sentinel-1 (Level-1 Single Look Complex) data from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31).
Acquisition period	2018-05-23 to 2018-09-08
Track	Ascending 90
Mode	IW
Incidence angle	52.2 °
Revisit	6 day
Reference point	Located on bedrock (69°13'48.37" N, 51°0'46.21" W)
Digital Elevation Model	Arctic DEM (10 m) (Porter et al., 2018)

Table 5. Line-of-sight (LOS) thaw-season ground surface displacement raster – year 2019.

Attribute	Description
File name	ILU_InSAR_LOS_surface_displacement_2019
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	2019 thaw-season ground surface displacement raster (unit m), provided in the radar line-of-sight (LOS) direction.
Derived from (raw data):	Sentinel-1 (Level-1 Single Look Complex) data from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31).
Acquisition period	2019-04-30 to 2019-09-15
Track	Ascending 90
Mode	IW
Incidence angle	52.2 °
Revisit	6 day
Reference point	Located on bedrock (69°13'48.37" N, 51°0'46.21" W)
Digital Elevation Model	Arctic DEM (10 m) (Porter et al., 2018)

2.2 2015-2019 maps of average seasonal ground surface displacement and long-term displacement rate

Table 6. Average seasonal ground surface displacement raster for the period 2015-2019.

Attribute	Description
File name	ILU_InSAR_avg_seas_displacement_raster_S_2015-2019
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site Extent	Ilulissat, West Greenland Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width Height Pixel size	680 pixels 885 pixels 10 m
Band(s)	1
Raster description	Average seasonal ground surface displacement raster for the period 2015-2019 (unit m). Representative of the ground surface deformations induced by freezing and thawing of the active layer. Provided for sedimentary deposits only (bedrock, freshwater and other non-vegetated areas excluded).
Derived from (raw data): Time period Retrieval technique	LOS thaw-season ground surface displacement rasters 2015 to 2019 Windowed linear least squares inversion

Table 7. Long-term ground surface displacement rate raster for the period 2015-2019.

Attribute	Description
File name	ILU_InSAR_long_term_displacement_rate_raster_R_2015-2019
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	Long-term ground surface displacement rate raster for the period 2015-2019 (unit m/yr). Related to changes in the permafrost table depth (permafrost degradation or aggradation). Provided for sedimentary deposits only (bedrock, freshwater and other non-vegetated areas excluded).
Derived from (raw data):	LOS thaw-season ground surface displacement rasters
Time period	2015 to 2019
Retrieval technique	Windowed linear least squares inversion

Table 8. Inversion's performance metrics – Coefficient of determination (R^2)

Attribute	Description
File name	ILU_InSAR_inversion_R2_coeff_2015-2019
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	Coefficient of determination, R^2 , computed for each raster pixel based on the values included in the inversion window. Provided for sedimentary deposits only (bedrock, freshwater and other non-vegetated areas excluded).

2.3 Land cover classification

Table 9. Non-vegetated land cover classes.

Attribute	Description
File name	ILU_RFC_land_cover_non_veget_classes_2016-2017
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	Non-vegetated classes derived by random forest classification of Sentinel-2 optical images.
Classes	0: Vegetated areas 1: Water (freshwater and seawater) 2: Gravel-dirt terrains 3: Bedrock 4: Built environment (buildings, roads, and other infrastructures) 5: Rocky-bare, sparsely vegetated areas (bushes growing on bedrock, cryptogam covers, etc.)
Derived from (raw data):	Sentinel-2 images (S2A_MSIL2A) from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31), and acquired during the summers of 2016 and 2017. The images were pre-processed to generate 10 bands resampled to 10 m spatial resolution (Lanaras et al., 2018).
Acquisition period	2016-08-13, 2017-08-08, and 2017-08-15
Classifier	Random Forest
Classification overall accuracy	82.51%
Color scale file	Color_scale_LC_non_veget.txt

Table 10. Vegetated land cover classes.

Attribute	Description
File name	ILU_RFC_land_cover_veget_classes_2016-2017
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site	Ilulissat, West Greenland
Extent	Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width	680 pixels
Height	885 pixels
Pixel size	10 m
Band(s)	1
Raster description	Vegetation classes derived by random forest classification of Sentinel-2 optical images.
Classes	0: Non-vegetated areas (bedrock, water, etc.) 1: EBDST (<i>Empetrum</i> and <i>Betula</i> -dominated dwarf shrub tundra) 2: BLDST (<i>Betula</i> -dominated low shrub tundra) 3: BSFB (<i>Betula</i> shrub-dominated frost boils) 4: SGCFB (shrub, graminoid and cryptogam-dominated frost boils) 5: BRSGT (bryophyte, rush and sedge-dominated graminoid tundra) 6: GGT (grass-dominated graminoid tundra) 7: TSGT (tussock sedge-dominated graminoid tundra) 8: Wetlands (permanently or seasonally flooded areas)
Derived from (raw data):	Sentinel-2 images (S2A.MSIL2A) from the European Space Agency Copernicus initiative (https://scihub.copernicus.eu/ , last accessed 2023-08-31), and acquired during the summers of 2016 and 2017. The images were pre-processed to obtain 10 bands resampled to 10 m spatial resolution (Lanaras et al., 2018).
Acquisition period	2016-08-13, 2017-08-08, and 2017-08-15
Classifier	Random Forest
Classification overall accuracy	82.51%
Color scale file	Color_scale_LC_veget.txt

2.4 Extrapolated active layer thicknesses

Table 11. Active layer thicknesses (2020-2021) extrapolated based on land cover classification.

Attribute	Description
File name	ILU_extrapolated_ALT_from_land_cover_2020-2021
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site Extent	Ilulissat, West Greenland Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width Height Pixel size	680 pixels 885 pixels 10 m
Band(s) No data value	1 -9999
Raster description	2020-2021 active layer thicknesses (unit cm) extrapolated based on land cover classification.
Derived from (raw data): Measurement period Extrapolation technique	Active layer thickness field measurements and land cover classification raster. Provided for sedimentary deposits only (bedrock, and other non-vegetated areas excluded). End of summer 2020 and 2021 Averaging per vegetation class

2.5 Frost susceptibility index

Table 12. Frost susceptibility index derived from average seasonal displacements and extrapolated active layer thicknesses.

Attribute	Description
File name	ILU_FSI_2015-2019
File format	TIF
Coordinate reference system	EPSG:3182 – GR96 UTM zone 22N
Site Extent	Ilulissat, West Greenland Upper left: 69°16'30.02" N, 51°9'25.60" W Lower left: 69°11'44.30" N, 51° 9'23.54" W Upper right: 69°16'30.28" N, 50°59' 5.76" W Lower right: 69°11'44.56" N, 50°59' 5.96" W
Width Height Pixel size	680 pixels 885 pixels 10 m
Band(s)	1
Raster description	Frost susceptibility index, representative of the average active layer ice content at the end of the thawing season.
Derived from (raw data): Study period	Average seasonal ground surface displacement raster, and extrapolated active layer thickness raster. Provided for sedimentary deposits only (bedrock, and other non-vegetated areas excluded). 2015-2019

3 GROUND TRUTH DATA SPECIFICATIONS

Table 13. Ground truth data file including active layer thickness measurements and vegetation surveys conducted in 2020 and 2021.

Attribute	Description
File name	ILU_ALT_measurements_Vegetation_surveys_2020-2021
File format	TAB-delimited txt
File description	Active layer thicknesses (ALT) were probed at the end of the summer in 2020 and/or 2021 at 377 sites. The surface characteristics and vegetation of those field sites were additionally described. Floristic surveys following the method of the Braun–Blanquet indices (Westhoff and Maarel, 1978; Braun-Blanquet, 2013) were conducted in 20 by 20, or 50 by 50 cm, quadrats at 84 of those locations, to assess the percent cover of the main plant functional types. The file contains the sites' coordinates, ALT measurements, surface characteristics, floristic compositions and vegetation classes.
Measurement period	End of summer 2020 and 2021

Table 14. Attributes of the ground truth data file.

Attribute name	Description
Site_ID	Name of the field site where the measurements were conducted
Loc_ID	Identification name given to the area where the measurements were conducted (ILU stands for Ilulissat)
Northing	Northing in UTM coordinates (zone 22)
Easting	Easting in UTM coordinates (zone 22)
Ref_syst	Coordinate reference system
EPSG	EPSG of the coordinate reference system
Soil_type	Main active layer soil type
ALT_2020	Active layer thickness (unit cm) at the end of the 2020 thawing season
ALT_2021	Active layer thickness (unit cm) at the end of the 2021 thawing season
Landform	Description of the geomorphological land form observed at the field site
Periglacial_features	Periglacial features observed at the field site
Transition	Boolean indicating if the field site was located at the transition between different vegetation units
Drainage	Description of surface drainage conditions
FlorSurv_Date	Date of the floristic survey
FlorSurv_area	Surface area (unit m ²) of the quadrat used to conduct the floristic survey
NV	Percent cover (unit %) of non-vegetated areas within the quadrat
S	Percent cover (unit %) of shrubs within the quadrat
F	Percent cover (unit %) of forbs within the quadrat
G	Percent cover (unit %) of graminoids within the quadrat
GG	Percent cover (unit %) of grasses within the quadrat
GS	Percent cover (unit %) of non-tussock sedges within the quadrat
GTS	Percent cover (unit %) of tussock sedges within the quadrat
GR	Percent cover (unit %) of rushes within the quadrat
L	Percent cover (unit %) of lichens within the quadrat
B	Percent cover (unit %) of bryophytes within the quadrat
P	Percent cover (unit %) of pteridophytes within the quadrat
Comments	Any comments related to the field measurements or site conditions
GT_V_name	Identification name of the vegetation class determined from the floristic composition: SLST (<i>Salix</i>-dominated low shrub tundra) EBDST (<i>Empetrum</i> and <i>Betula</i>-dominated dwarf shrub tundra) BLDST (<i>Betula</i>-dominated low shrub tundra) BSBFB (<i>Betula</i> shrub-dominated, partially bare frost boils) BSVFB (<i>Betula</i> shrub-dominated vegetated frost boils) SGCBFB (shrub, graminoid and cryptogam-dominated, partially bare frost boils) SGCVFB (shrub, graminoid and cryptogam-dominated vegetated frost boils) LBFB (lichen-dominated, partially bare frost boils) LVFB (lichen-dominated vegetated frost boils) TSGT (tussock sedge-dominated graminoid tundra) BRSGT (bryophyte, rush and sedge-dominated graminoid tundra) GGT (grass-dominated graminoid tundra) SGBT (shrub, graminoid and bryophyte-dominated tundra)

Attribute name	Description
Fines	Boolean indicating the presence/absence of fine-grained sediments in the active layer
LONGITUDE	WGS84 longitude of the field site
LATITUDE	WGS84 latitude of the field site

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